FOR A NOETHERIAN RING THE NAIVE COMPLETION FUNCTOR IS LEFT ADJOINT TO THE INCLUSION OF DERIVED COMPLETE OBJECTS

THE STACKS PROJECT AUTHORS

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Editorial note. The counted proof span contains a displayed three-term chain of equalities; it is delivered verbatim and the extra parentheses around 'A/(f_1^n, \ldots, f_r^n))' in the source are reproduced as they stand. The four references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- The Koszul complex description of derived completion and the construction of the inclusion functor D_comp(A) -> D(A). invoked at proof lines 27229-27230 and referred to at statement lines 27222-27223. Provides the adjunction framework and the reduced statement to be proved.
- The pro-objects given by the Koszul complexes and by the quotients by the powers of the ideal are isomorphic. invoked at proof lines 27236-27238. Transports the computation between the two presentations of I.
- Tensor products and R lim respect isomorphism of pro-objects. invoked at proof lines 27240-27241. Concludes that the comparison map is an isomorphism.

Source reference. Original lemma 26154: sources/stacks-more-algebra/pinned-original-source.tex, source lines 26154–26154. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 26388: sources/stacks-more-algebra/pinned-original-source.tex, source lines 26388–26388. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 27172: sources/stacks-more-algebra/pinned-original-source.tex, source lines 27172–27172. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

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Source reference. Original lemma 23939: sources/stacks-more-algebra/pinned-original-source.tex, source lines 23939–23939. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Proposition 0.2. *Let A be a Noetherian ring. Let $I \subset A$ be an ideal. The functor which sends $K \in D(A)$ to the derived limit $K' = R\lim(K \otimes^{\mathbf{L}}_A A/I^n)$ is the left adjoint to the inclusion functor $D_{comp}(A) \to D(A)$ constructed in Lemma 26154.*

Proof. Say $(f_1, \ldots, f_r) = I$ and let $K_n^\bullet$ be the Koszul complex with respect to $f_1^n, \ldots, f_r^n$. By Lemma 26388 it suffices to prove that

$$R\lim(K \otimes^{\mathbf{L}}_A K_n^\bullet) = R\lim(K \otimes^{\mathbf{L}}_A A/(f_1^n, \ldots, f_r^n)) = R\lim(K \otimes^{\mathbf{L}}_A A/I^n).$$

By Lemma 27172 the pro-objects $\{K_n^\bullet\}$ and $\{A/(f_1^n, \ldots, f_r^n)\}$ of $D(A)$ are isomorphic. It is clear that the pro-objects $\{A/(f_1^n, \ldots, f_r^n)\}$ and $\{A/I^n\}$ are isomorphic. Thus the map from left to right is an isomorphism by Lemma 23939. □